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Campus Information Retrieval Assistant Using Small Language Model

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INTRODUCTION

- Universities generate vast amounts of information across multiple sources such as websites, PDFs, and notices
- Students and stakeholders often struggle to access relevant information quickly
- Traditional search methods are inefficient and time-consuming
- AI-based assistants can significantly improve information accessibility
- This project introduces a campus-specific intelligent assistant

PROBLEM STATEMENT

- Campus information is highly fragmented across different platforms
- Users spend significant time searching for simple queries
- Existing chatbot systems rely on cloud-based LLMs which cause:
  - High operational costs
  - Privacy and data security risks
  - Hallucinated or inaccurate responses
- Lack of systems prioritizing institutional data sources

Therefore, a reliable, privacy-focused, campus-specific solution is required

OBJECTIVES

- Design a Small Language Model (SLM)-based assistant
- Provide accurate and fast responses
- Minimise hallucinations using RAG
- Enable on-premise deployment
- Prioritise institutional data sources
- Support natural language queries

PROPOSED SOLUTION

Develop a Campus Information Retrieval Assistant using:

- Small Language Model (SLM)
- Retrieval-Augmented Generation (RAG)

System Workflow:

- User submits query
- System retrieves relevant data
- Generates accurate, context-based response

Data Priority:

- Institutional PDFs
- FAQs
- External sources (if needed)

CIRA

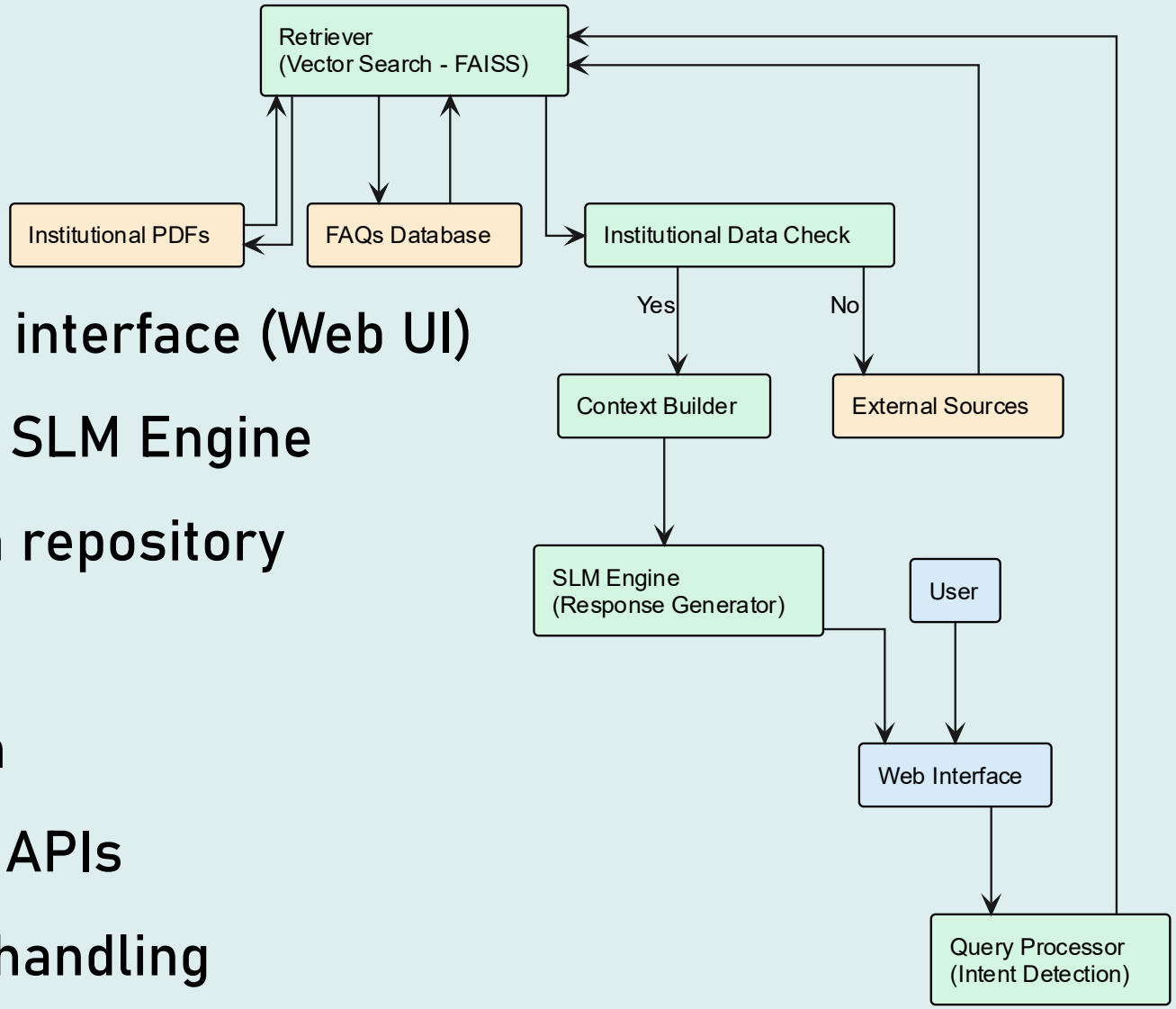
System Architecture

Three-Layer Architecture:

- Presentation Layer: Chat interface (Web UI)
- Processing Layer: RAG + SLM Engine
- Data Layer: Campus data repository

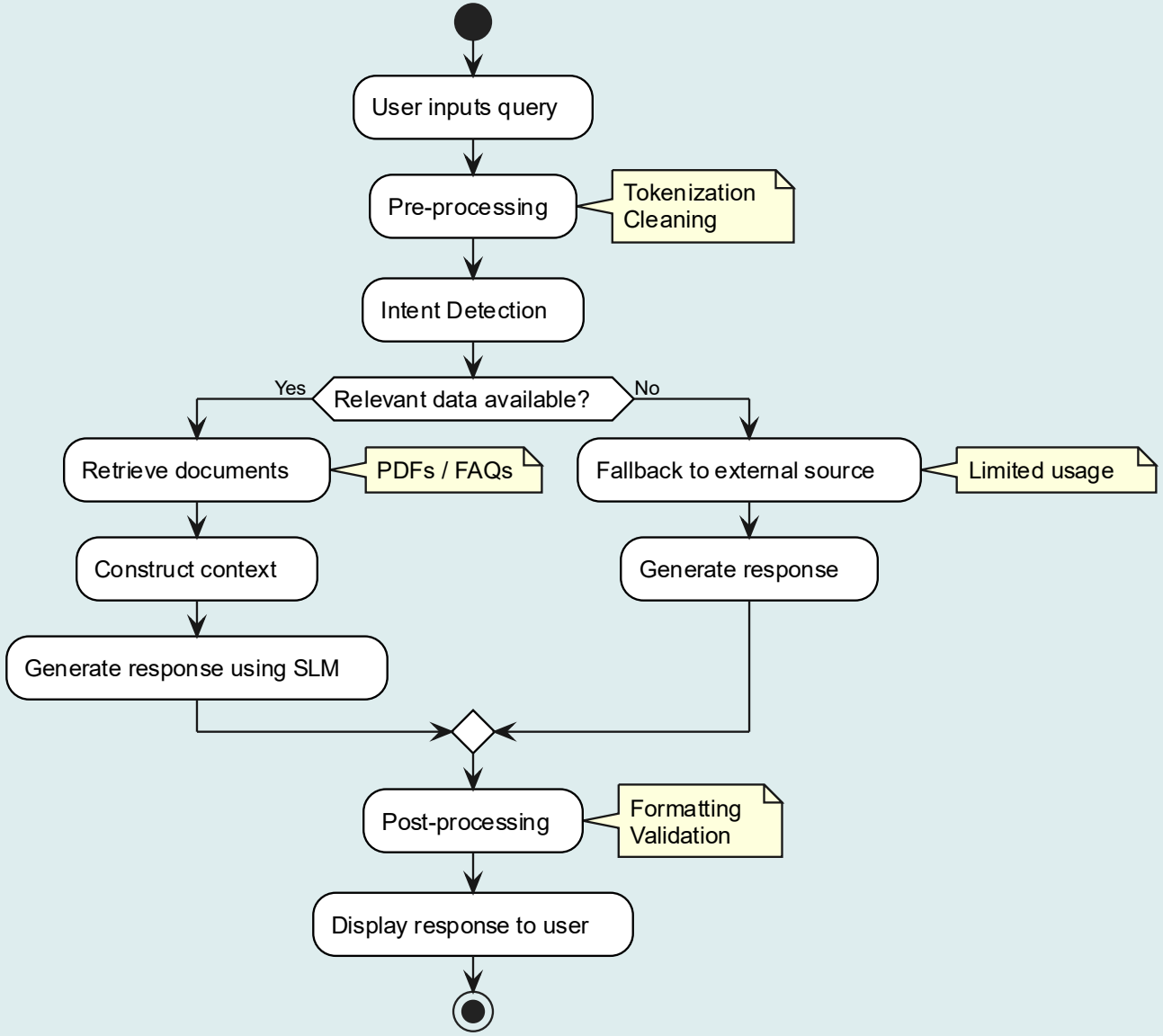
Key Features:

- Fully on-premise system
- No dependency on cloud APIs
- Secure and private data handling



Working Principle

1. User inputs query
2. System understands intent
3. Relevant documents retrieved
4. Context is formed
5. SLM generates final response
6. Response delivered to user



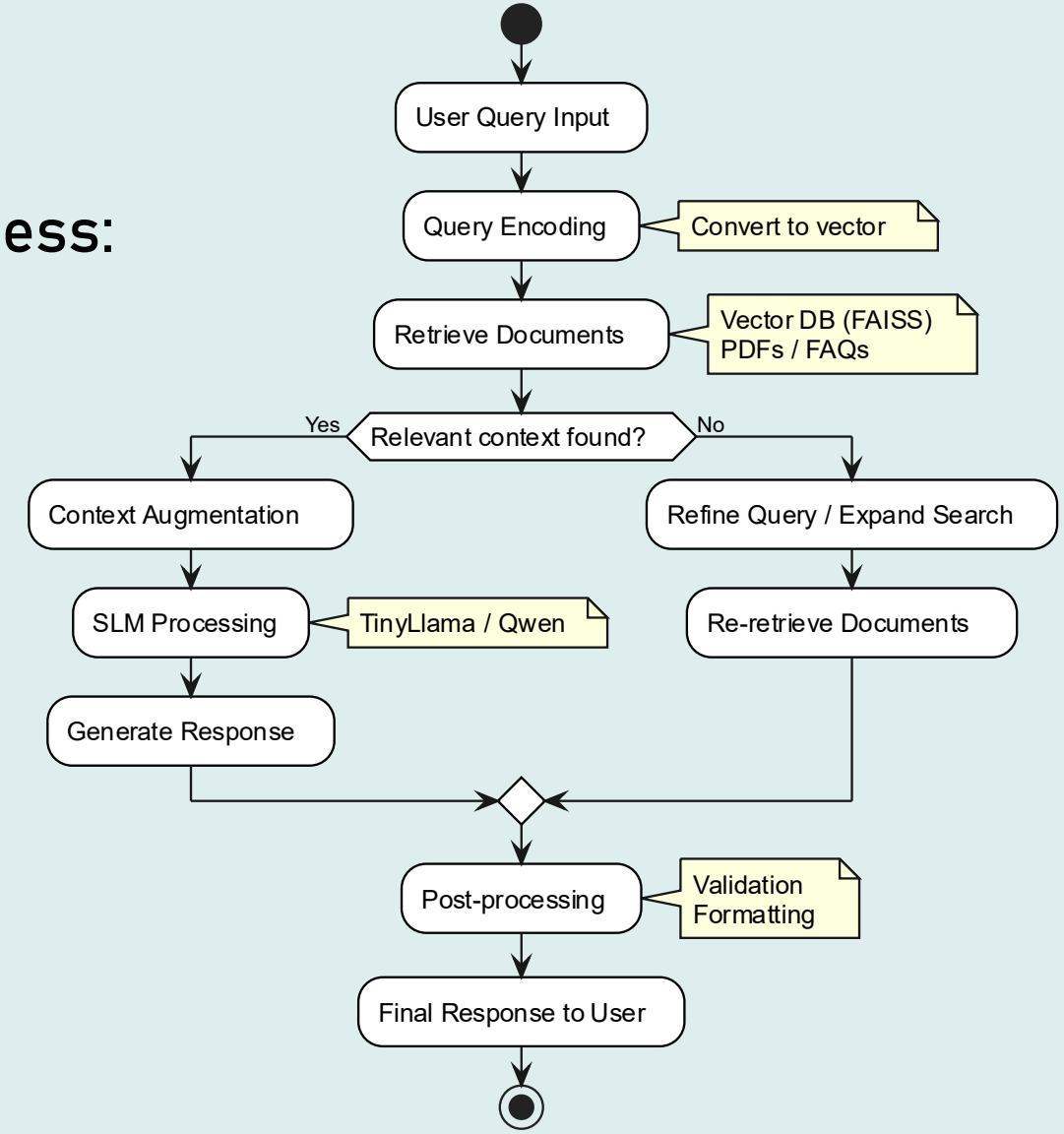
Methodology

Retrieval-Augmented Generation Process:

- Retrieve relevant documents
- Augment query with context
- Generate response using SLM

Advantages:

- Reduces hallucinations
- Ensures factual accuracy
- Uses trusted institutional data



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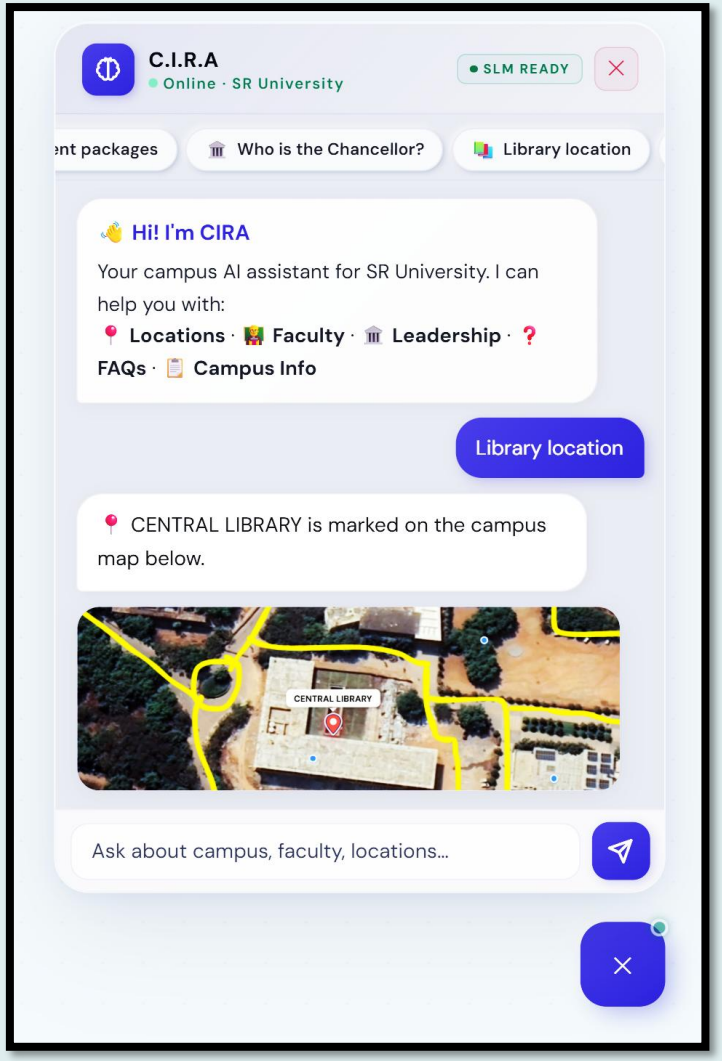
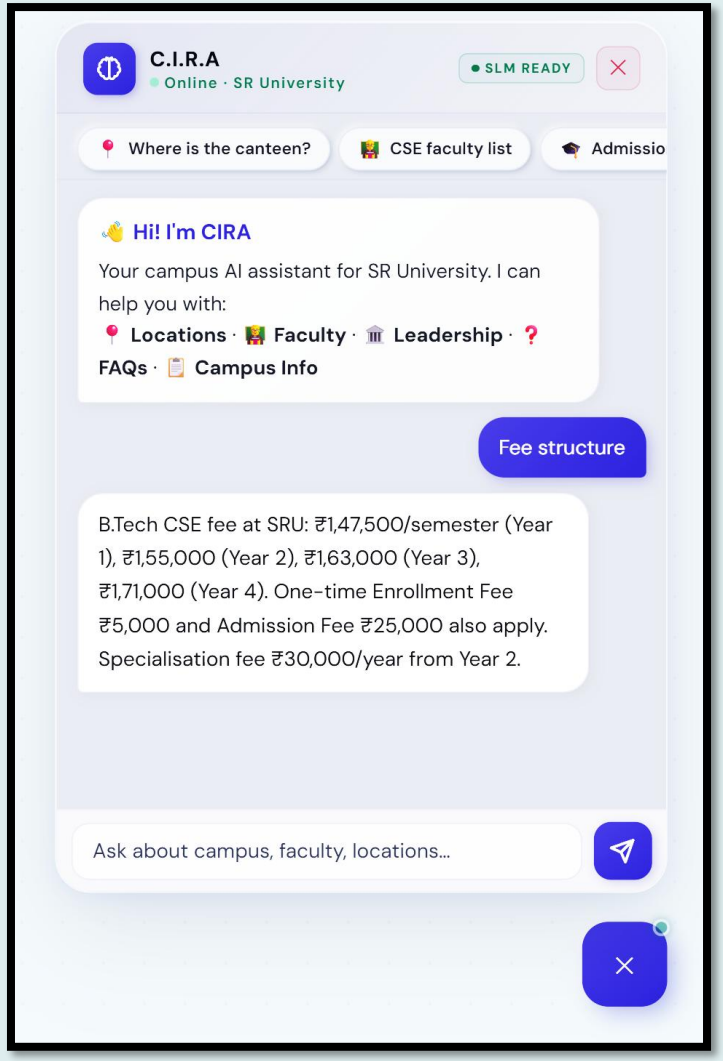
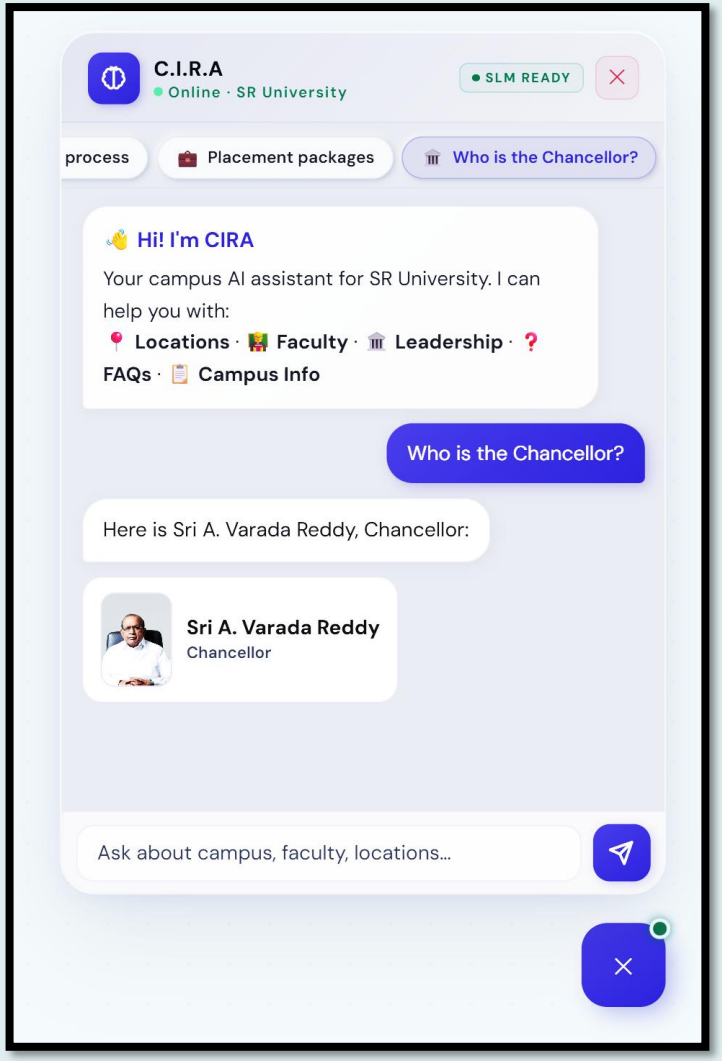
Results







Results & Advantages

- High accuracy due to grounded responses

- Zero cloud dependency ensures privacy

- Reduced operational cost

- Faster response time

- Scalable for university-wide deployment

Conclusion

- Successfully developed a campus-specific intelligent assistant
- Provides accurate, reliable, and context-aware responses
- Suitable for real-world deployment

Future Scope:

- Voice-based interaction
- Mobile application integration
- Multi-language support
- Integration with university ERP systems